Molecular Geometry

- Atom of greater electronegativity will attract electron density to a larger extent, developing a slightly negative charge.

Bond length - the avg distance between the nuclei of a pair of bonding atoms.

Bond angle - the angle formed between 3 bonding atoms.

1. Determine # of VÉ
2. Determine Electronegativity
3. Lewis diagram
4. Bonding / Non Bonding angles
5. Shape?

- going towards you
- going away from you
- 3, 6, 9 o'clock

VSPER theory

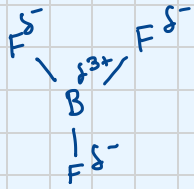
- Valence Shell Electron Pair repulsion theory
- Electron pairs repel eachother as much as they can in 3-dimentional space.
- Depends on the total # of electron pairs, both bonding & non-bonding.

1. ***Linear*** : $BeCl_2$

Central atom: Be
bonding electron pairs: 2
non-bonding electron pairs: 0
Bond Angle: 180°

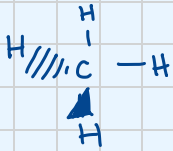
2. ***Trigonal Planar***: BF_3

Central atom: B
bonding electron pairs: 3
non-bonding electron pairs: 0
Bond Angle: 120°



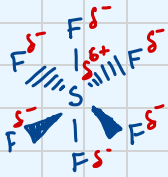
3. ***Tetrahedral***: CH_4

Central atom: C
bonding electron pairs: 4
non-bonding electron pairs: 0
Bond Angle: 109.5°



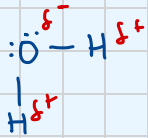
4. ***Octahedral***: SF_6

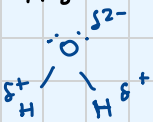
Central atom: S
bonding electron pairs: 6
non-bonding electron pairs: 0
Bond Angle: 90°



5. ***Angular***: H_2O

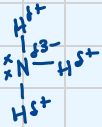
Central atom: O
bonding electron pairs: 2
non- bonding electron pairs: 2
Bond Angle: 104.5°

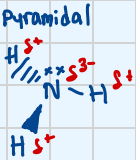
1. Total: 8 vé
2. ΔEN: 1.24 P.C
3. 
4. Central atom: O
   Bonding: 2
   non-bonding: 2
   angle: 104.5°
5. Angular ☆



6. ***Pyramidal***: NH_3

Central atom: N
bonding electron pairs: 3
non-bonding electron pairs: 1
Bond Angle: 107.5°

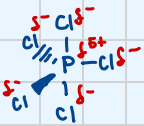
1. Total: 8 vé
2. ΔEN: 0.84 P.C.
3. 
4. Central atom: N
   Bonding: 3
   non-bonding: 1
   angle: 107.5
5. Pyramidal



7. ***Trigonal Bipyramid***: PCl_5

Central atom: P
bonding electron pairs: 5
non-bonding electron pairs: 0
Bond Angles: 90° & 120°

2. ΔEN: 0.97 P.C
4. Central atom: P
   Bonding: 5
   non-bonding: 0
   angle: 90° & 120°
5. Trigonal Bypyramid



Explaining Properties of molecular crystals

1. liquids/solids do not conduct electricity at any extent.

Atoms are bonded through very strong covalent bonds, consisting of 2 electrons that are shared between two ajacent atoms. The electrons are locked in place and not free to move. This results in the compound cannot conduct electricity.

2. Many exist as gases at room temp/atmospheric pressure. Many solids/liquids are Volatile (easily evapourated).

Molecular solids/liquids are held together by very weak VDW forces. This allows them to seperate very easily, thus making them volitile.

3. Melting point of solids are relatively low

Forces of attraction between molecules is very weak, very little energy is required to seperate these particles.

4. Boiling point of liquids are relatively low.

Same as #3, but liquid particles seperate to form a gas.

5. solids are generally soft and waxy.

Since they're held together by very weak forces of attraction, it is easy to cause those particles to move over one another when applying force.

6. Large amounts of energy is required to decompose the substance chemically into simpler substances.

A great deal of energy is required to decompose molecules because the intramolecular force must be very strong.

1. Why does water turn into steam at 100°C but does not decompose into H_2 & O_2 atoms at 100°C?

Water turns into steam at 100°C because of intermolecular forces between water molecules break allowing gas to escape, a physical change. On the contrary, decomposing into H_2 & O_2 is a chemical change which requires breaking strong covalent bonds. It takes a lot more energy to break up the molecule into atoms.

2. Why is dry ice more volatile than ice?

Dry ice will turn into gas without melting into a liquid (weak dispersion force). Ice melts at 0°C requiring more heat to evapourate (strong hydrogen bonds). Because it turns into gas at a lower temp, CO_2 is more volatile than H_2O.

3. Pentane's Boiling point/melting point is higher than neopentane, explain why in terms of molecular bonding.

Same formula, different structure/composition.

pentane is a straight chain molecule. Since the molecules can align closely together because of its dispersion forces, it needs more energy to seperate them, leading to a high boiling point.

Neopentane is a symmetrical molecule, making it harder for molecules to interact strongly with eachother. Weaker dispersion forces, lower boiling point since molecules seperate easily.

neopentane has a higher melting point and pentane has a higher boiling point. This is because in their solid forms neo is more closely compact than normal. After becoming liquids, normal has greater forces of attraction than neo now.

4.

Alcohols

Structural Diagram

R — O — H

They are classified as derivatives of water, having the functional group of hydroxyl (−OH)

Hydroxyl groups

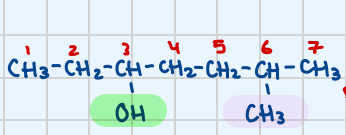
CH_3-OH	methanol
CH_3CH_2-OH	ethanol
$CH_3CH_2CH_2-OH$	propan-1-ol
$CH_3-\underset{\underset{OH}{\mid}}{CH}-CH_3$	propan-2-ol

Naming

1. Identify longest chain which has the hydroxyl group attached to it.
2. Ending is changed to "ol", number using priority chart.

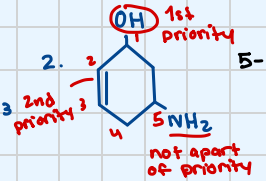
Examples

1.



6-methylheptan-3-ol

2.

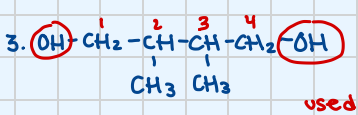


5-aminocyclohex-2-enol

don't forget cyclo

assumed at 1 so no need to write en-1-ol.

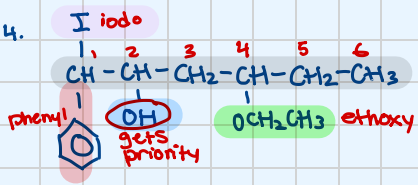
3.



2,3-dimethyl butanediol

used when theres two alcohol attachements.

4.



4-ethoxy-1-iodo-1-phenylhexan-2-ol

Properties

Molecular polarity: Very polar

Intermolecular bond type: Hydrogen bonds

Strength of Intermolecular bond: Very strong

Solubility: The hydroxyl group adds polarity to the molecule while the rest remains non-polar. As chain becomes larger, solubility decreases.

Volatility: Slightly Volitile

Odour: Slight odour

Boiling Point: Much higher compared to alkanes & ethers.

Uses

- Alcoholic Beverages
- Disinfectants
- Moisturizers
- Chocolates

Comparing polarity of alcohol & Ethers

Ethers are slightly more polar than alkanes due to electron density supplied by oxygen atom. Alcohol's are more polar because of the hydroxyl group (H-bonding). water is even more polar because of the two H-bonds.

bp/mp increases

FOA between molecules increase

Compound	Structure	Polarity	Boiling Point (°C)
ethane	$CH_3 — CH_3$	none	-89
methoxymethane	$CH_3 — O — CH_3$	low	-23
ethanol	$CH_3 — CH_2 — O — H$	moderate	78.5
water	$H — O — H$	extremely high	100